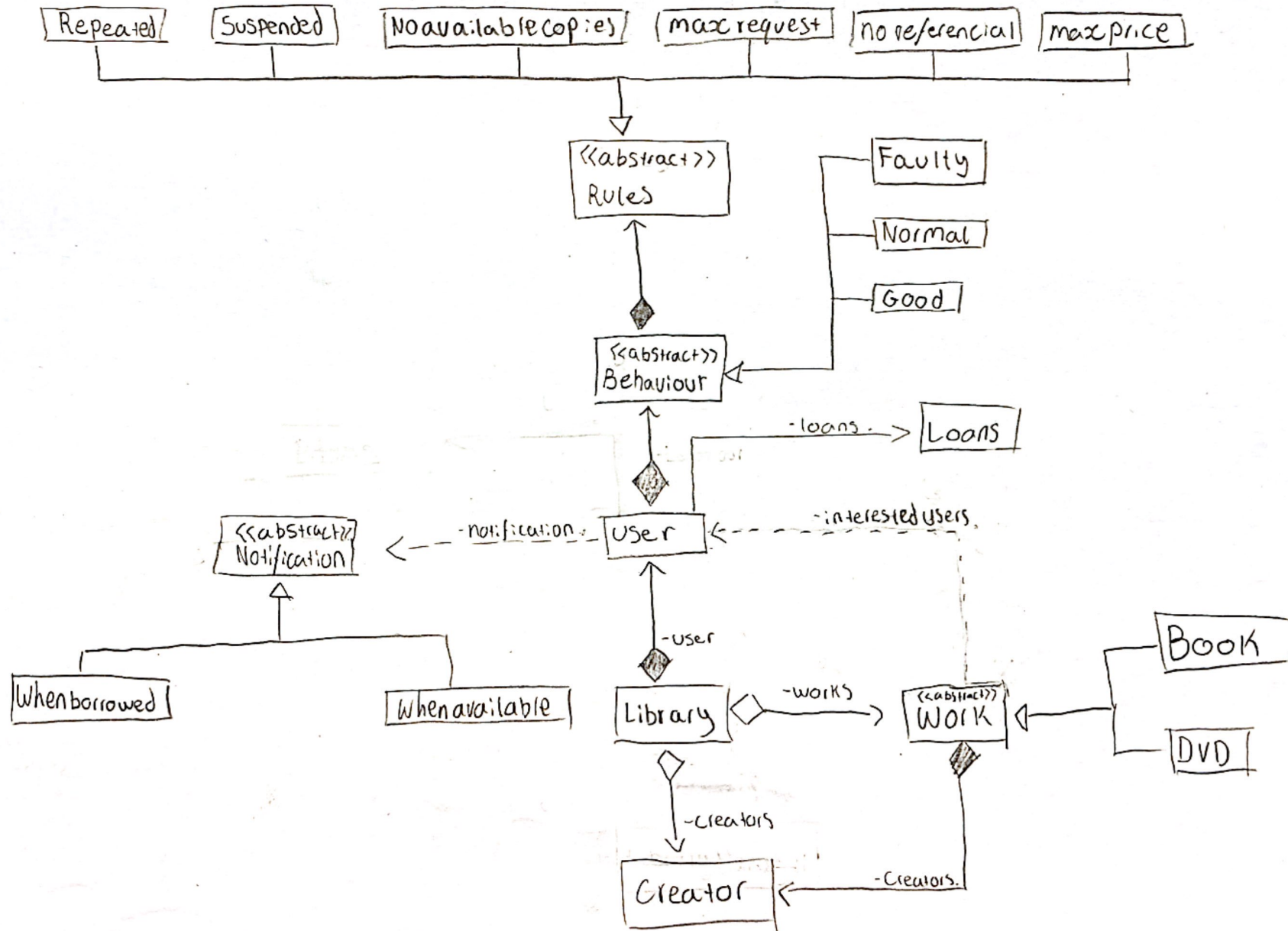


Declaro por minha honra que este diagrama foi realizado apenas pelos elementos que constituem o grupo da Projeto. Margarida Guedes: 1111442
Declaro por minha honra que este diagrama foi realizado apenas pelos elementos que constituem o grupo do Projeto. Marta Mendes: 11114044 grupo 087



Declaro por minha honra que este diagrama foi realizado apenas pelos elementos que constituem o grupo do projeto Margarida Guedes
 Declaro por minha honra que este diagrama foi realizado apenas pelos elementos que constituem o grupo do projeto Margarida Guedes 151114497
 grupo 087

```

Library Manager
- library : Library
+ save() : void
+ save as (filename: string) : void
+ load (filename: string) : void
+ import file (filename: string) : void
    
```

```

Loan
- due date : int
- returned date : int
+ mark returned (date: int) : void
+ is late (current date: int) : boolean
+ can borrow (u: user, w: work) : boolean
    
```

```

Work <<abstract>>
- <<final>> id : int
- title : string
- price : int
- category : string
- available copies : int
- total copies : int
- interested users : List<user>
- <<final>> type : string
+ borrow copy (u: user) : void
+ return copy (u: user) : void
+ change available copies (n-copies: int) : void
+ has available copies() : boolean
+ add interested user (u: user) : void
+ notify interested users() : void
    
```

```

Notification <<abstract>>
- read : boolean
- message : string
+ send() : void
    
```

```

When Borrowed
+ send() : void
    
```

```

When available
+ send() : void
    
```

```

DVD
- IGAC : string
+ get Director() : creator
    
```

```

Book
- ISBN : string
+ get Authors() : List<creators>
    
```

```

Creator
- name : string
- works : List<works>
    
```

```

Library
- current Date : int
- user : MAP<Integer, user>
- works : MAP<Integer, work>
- creators : MAP<string, creator>
- last user ID : int
- last work ID : int
+ import file (filename: string) : void
+ add user (u: user) : void
+ add work (w: work) : void
+ add creator (c: creator) : void
+ remove user (u: user) : void
+ remove work (w: work) : void
+ remove creator (c: creator) : void
+ get current date() : int
+ get user by Id (id: int) : user
+ get works by Id (id: int) : work
+ get creator by name (name: string) : creator
+ advance date() : void
+ search works (term: string) : void
    
```

Declaro por minha honra que este diagrama foi realizado apenas pelos elementos que constituem o grupo do Projeto. Margarida Gueles 15111444
Declaro por minha honra que este diagrama foi realizado apenas pelos elementos que constituem o grupo do projeto. Marfa Mendes 151114049

Grupo 087

Rules <<abstract>>

- name: string
+ check(u: user, w: work): boolean
+ is applicable(u: user, w: work): boolean

Repeated

+ check(u: user, w: work): boolean

Suspended

+ check(u: user, w: work): boolean

No available copies

+ check(u: user, w: work): boolean

Max request

+ check(u: user, w: work): boolean

No referencial

+ check(u: user, w: work): boolean

max price

+ check(u: user, w: work): boolean
+ is applicable(u: user, w: work): boolean

Behaviour <<abstract>>

- name: string
- rules: List<Rules>

+ can request(u: user, w: work): boolean
+ request loan duration(u: user, w: work): int

Faulty

+ request loan duration(u: user, w: work): int

Normal

+ request loan duration(u: user, w: work): int

Good

+ request loan duration(u: user, w: work): int

User

- id: int
- name: string
- email: string
- status: string
- behaviour: Behaviour
- loans: List<Loan>
- notification: List<notification>
- borrowed works: List<work>
- fines: int
+ request work(w: work): void
+ return work(w: work): void
+ pay fine(fine: int): void
+ add notification(n: notification): void
+ make read(n: notification): void